

Folding the Future of Packaging

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Replacement of Styrofoam to a Paper-Based Material for Shipment



Agenda

Background

Design Process

Engineered Cardboard Solution

Our Vision

Conclusion

Problems with Styrofoam in Packaging

- **Environmental Impact:**
Non-biodegradable, takes centuries to decompose, major landfill and ocean pollutant.
- **Health & Recycling Issues:**
Contains toxic chemicals, difficult and costly to recycle, contributes to microplastic pollution.
- **California & US Regulations:**
Growing bans on Styrofoam (e.g., California AB 1201, local ordinances), push for sustainable alternatives in packaging.
- Opportunity for brands to adopt eco-friendly materials and align with evolving packaging standards.





Competitiveness and Cost

Styrofoam:

- A little cheaper than cardboard
- Styrofoam is not reusable
- Slower production speed
- Very customizable
- Lots of production facilities

Engineered Cardboard

- Cost-effective material
 - Recyclable and upcyclable
 - Faster production speed
 - Moderately customizable
 - Wide variety of available materials
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Project Ask – Sustainable Packaging

- **Goal:** Replace EPS with a paper-based, cost-effective alternative.
- **Challenge:** Current paper solutions lack durability, deform under pressure, and impact product presentation.
- **Constraints:**
 - Fully paper-based materials (including cardboard).
 - Must withstand 20 lb product drop test (6 drops from 70 cm).
 - Competitive cost with innovative structural solutions.



Selecting the Approach

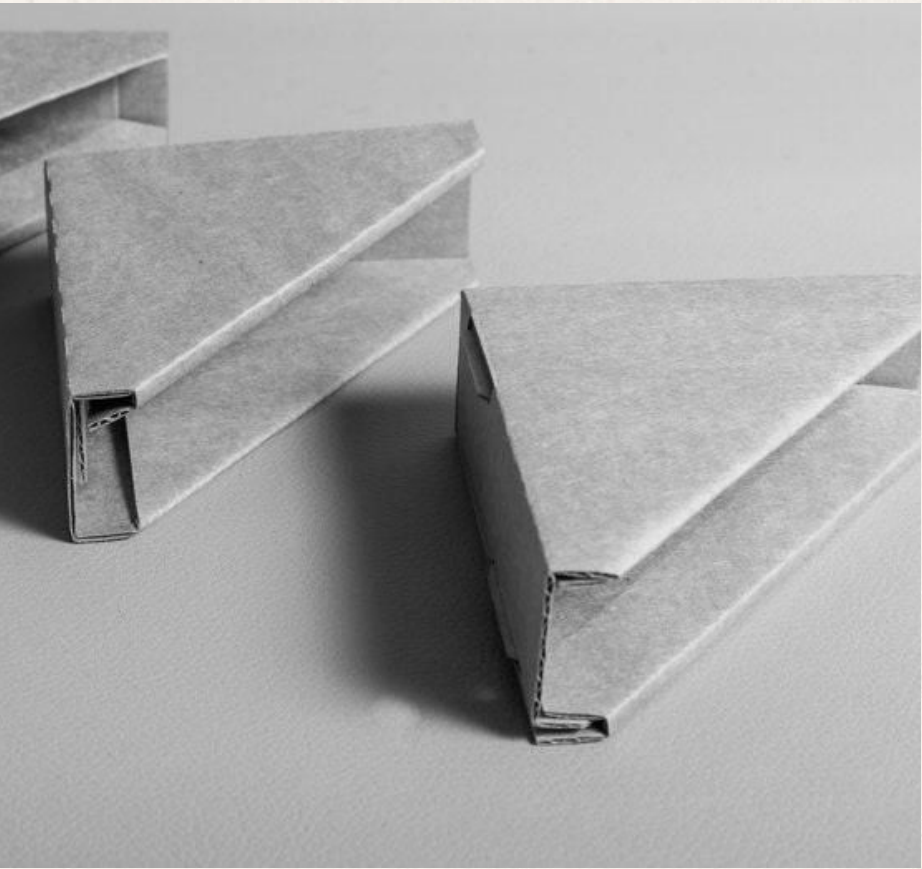
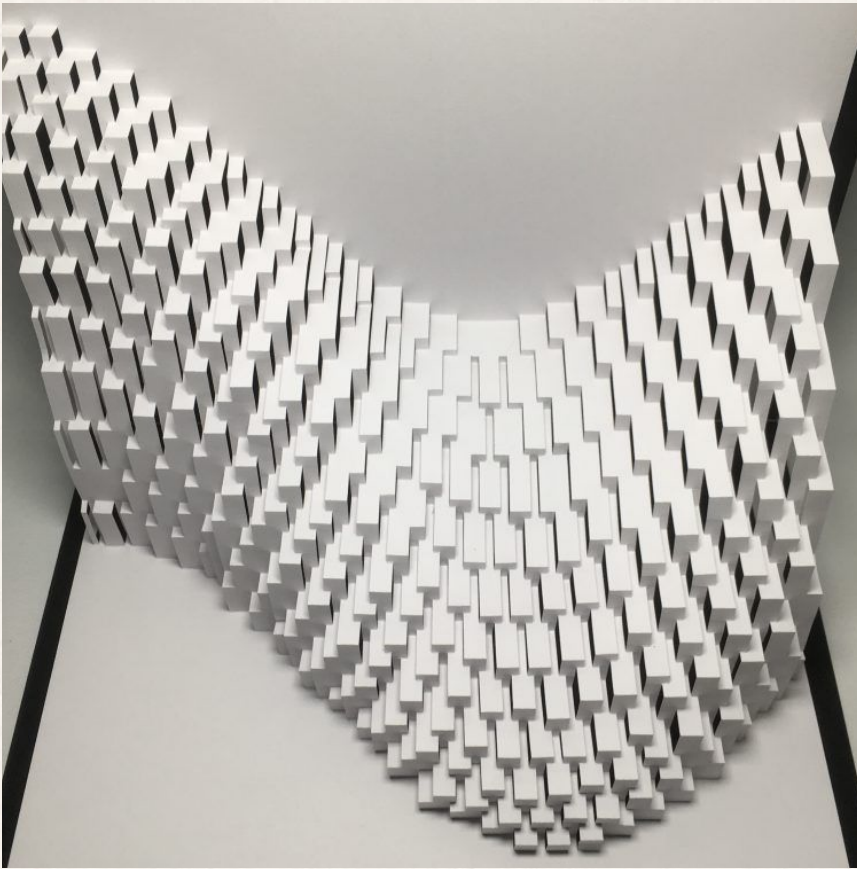
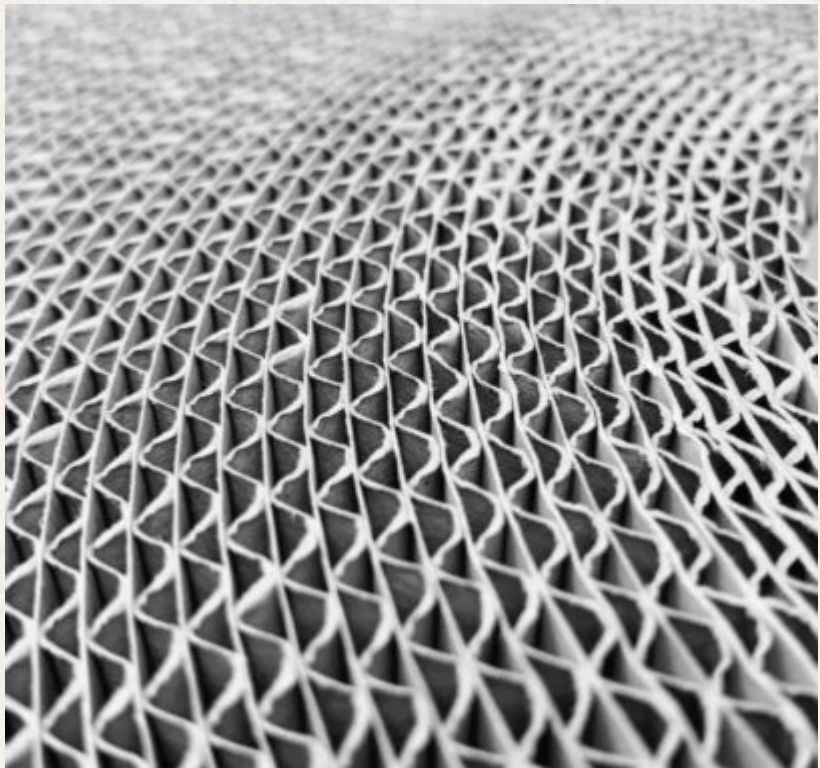
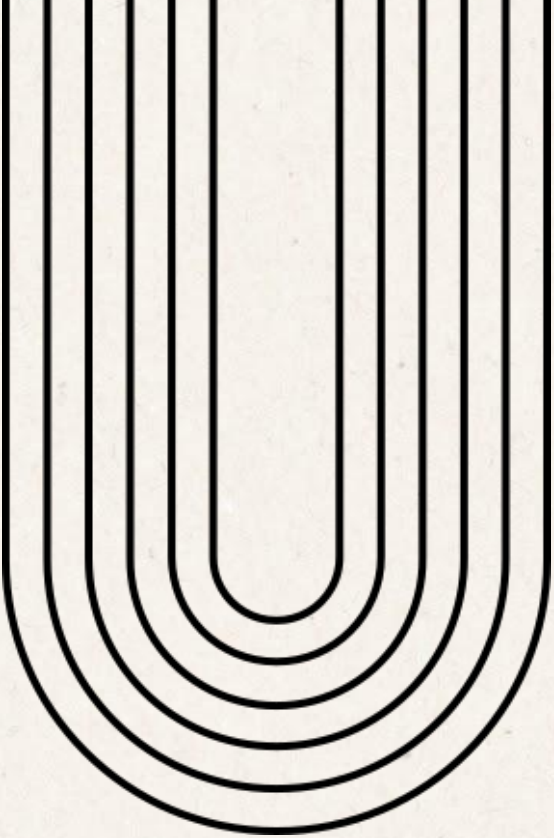
After attempting several ideas we settled on a mix of cardboard solutions

**Cardboard Pulp
Molding**

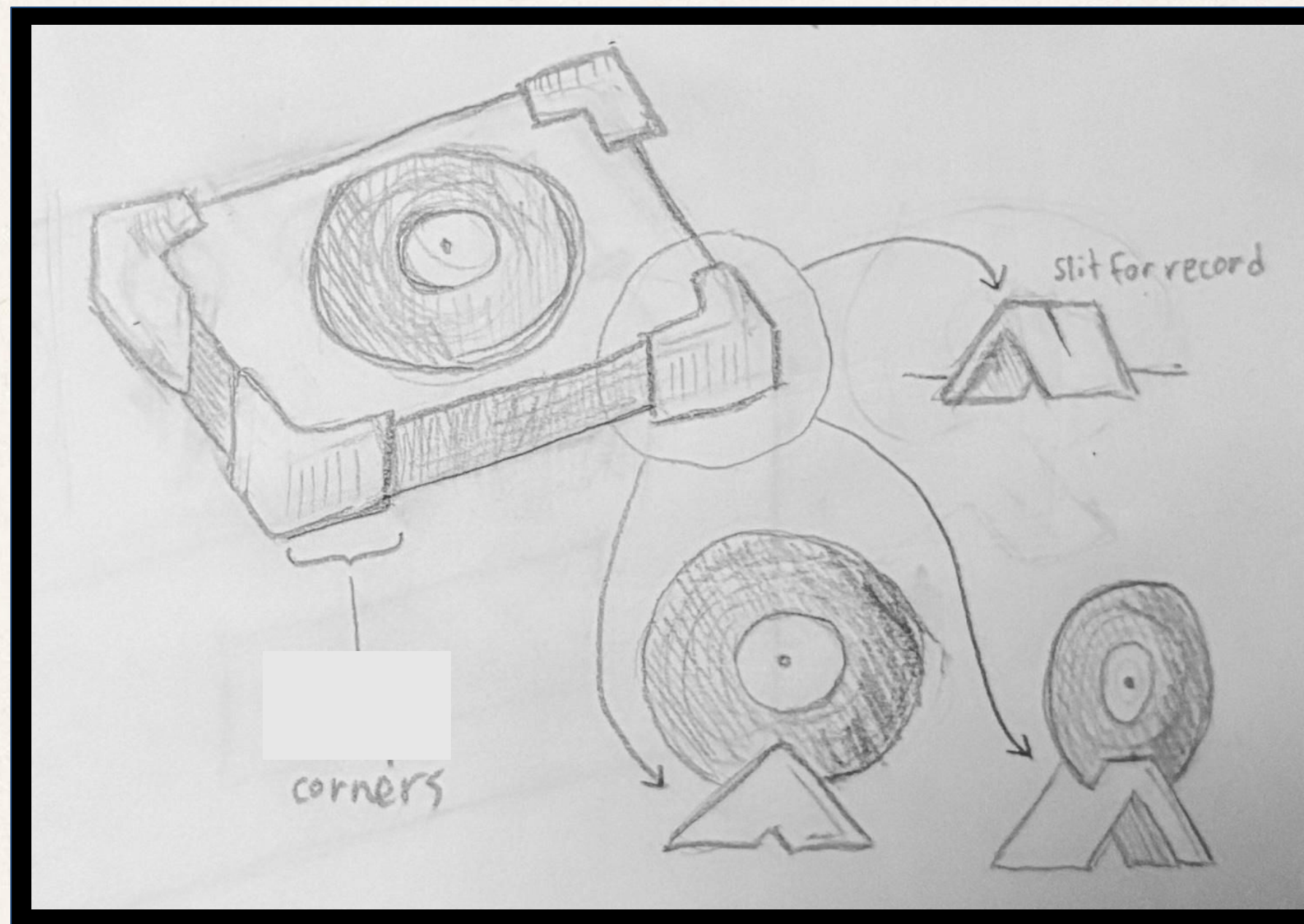
**Corrugation
Manipulation**

Kirigami

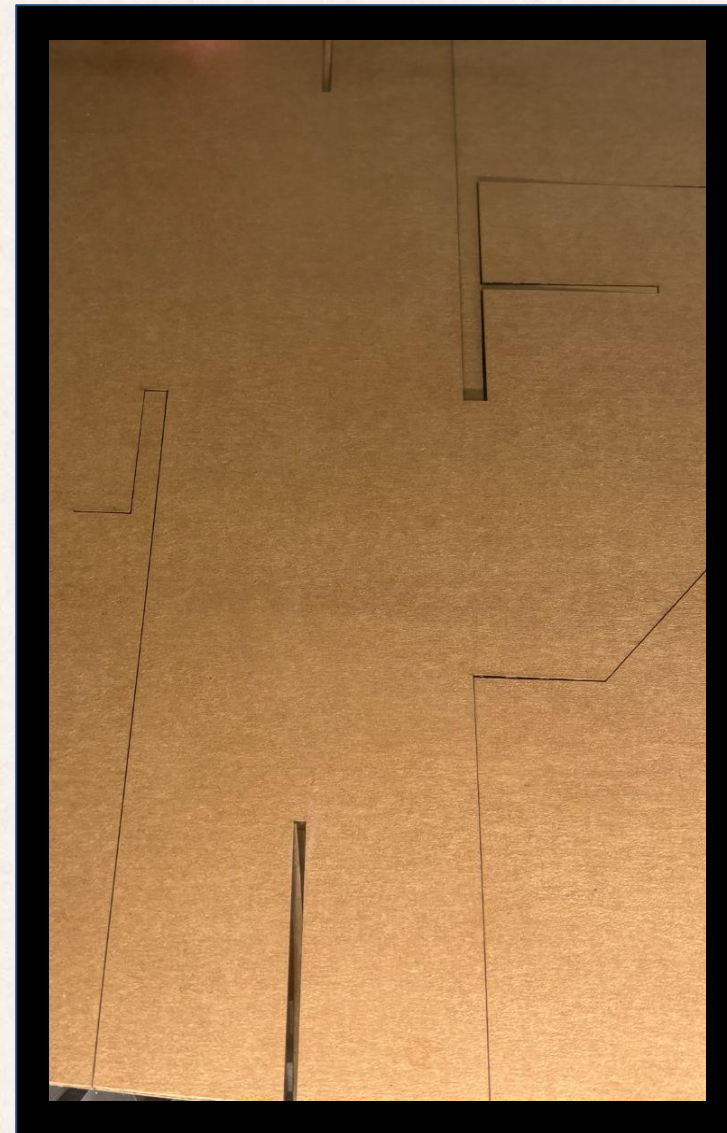
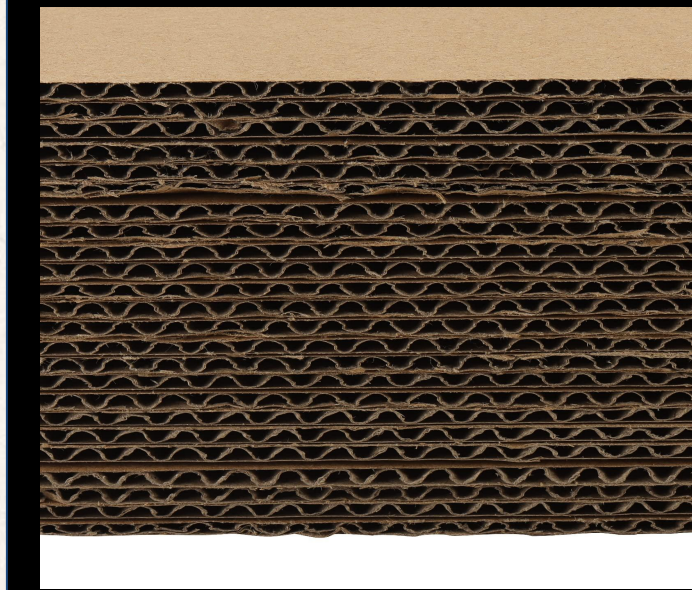
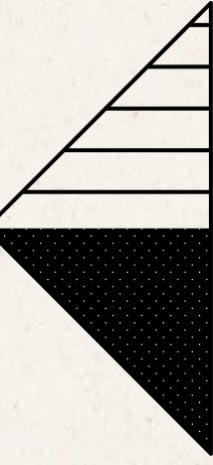
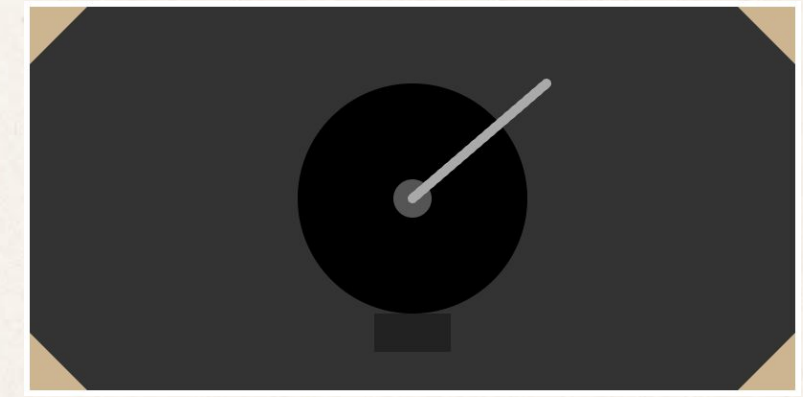
Folded Cardboard



Multi-Use for the Packaging



Corner Development Process



Engineered Cardboard Process

- After feedback that molded pulp was **not production-feasible**, we pivoted to **engineered cardboard**.
- We designed packaging elements using standard **corrugated** cardboard.
- We prototyped foldable **kirigami-style corners**, **stacked cross-sections**, **woven mesh**, and **gap-filling barriers**.



Our First Molded Pulp Prototype



Paper Corner Concept w/ Record



First Corner Version



Mesh Woven Cardboard



Stacked Cross Sections



What our Process Revealed

- Corner protectors alone were **insufficient** to fully protect the turntable.
- Added cardboard barriers with **corrugated cross-sections** against the turntable sides for stronger lateral protection.
- To avoid damaging the **plastic casing**, we rotated barriers so that the smooth cardboard surface touched the turntable instead of the rigid cross-section.
- Added **flexible woven cardboard mesh** to cushion the top and bottom



Testing Process

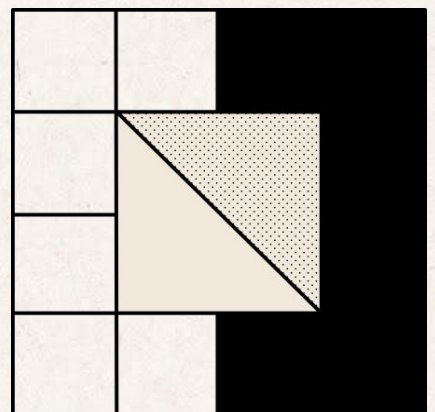
- Used the conservation of energy principle ($m_1gh_1 = m_2gh_2$) to determine equivalent drop heights for testing.
- Simulated dropping a **20 lb (9.1 kg)** turntable from **70 cm** by drop-testing our **5.73 lb (2.6 kg)** turntable from **8.0 ft (2.44 m)**.
 - Conducted one full round of drop tests from 70 cm, dropping each side of the packaged turntable.
 - Conducted two full rounds of drop tests from 8.0 ft (2.44 m).
- Testing was **halted** after two rounds at 8 ft due to packaging and turntable damage; unable to complete the full six-round test protocol.



*Gaven Measuring
the 8ft drop*

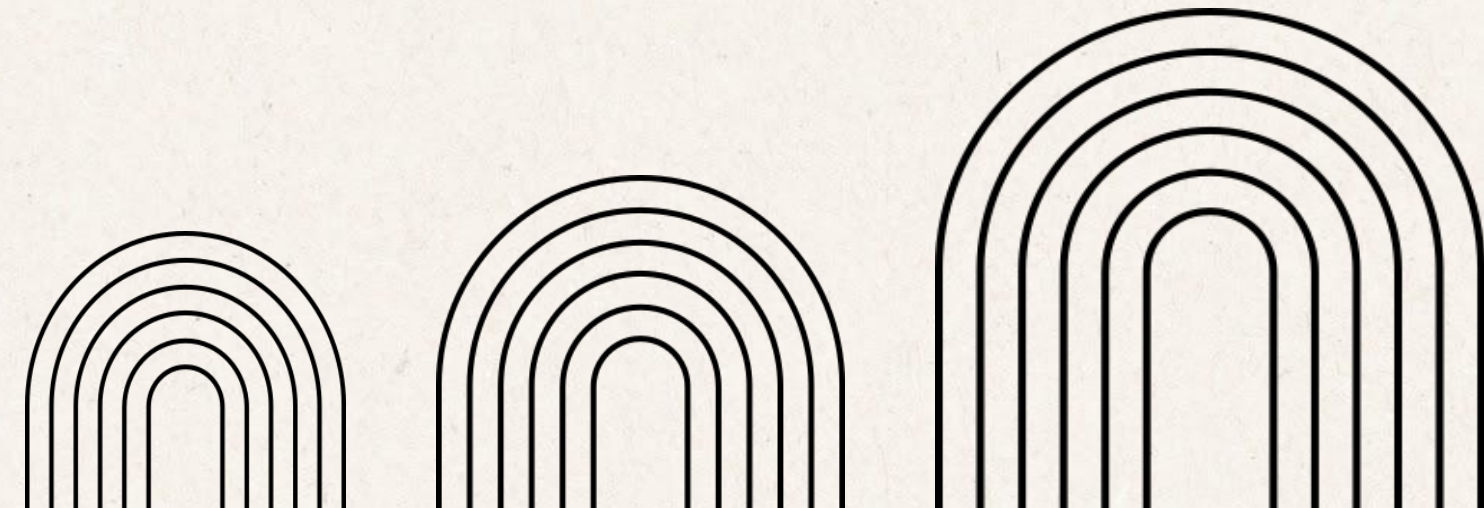
Testing Results

- Initial tests conducted from a height of 70 cm resulted in one of the plastic hinges chipping and no further damage.
- Tests conducted from 8.0 ft (2.44 m) led to severe damage of the turntable after two rounds
 - The top and bottom of the turntable were much more resilient to impact, resulted in little to no damage
 - Protected using cardboard mesh
 - The sides were the most vulnerable to impact
 - Typically resulted in chipping of the plastic casing
 - Eventually a variety of other components of the turntable became damaged
 - Protected by cardboard barriers and cardboard corners



Design Learning Outcomes

- Weaknesses:
 - Insufficient shock absorption along the sides of the record player.
 - Too much movement space inside the box.
 - Back hinges were especially vulnerable to impact damage.
- Strengths:
 - Excellent shock absorption from the woven cardboard mesh.
 - High versatility and adaptability of the kirigami cardboard corners.



Dual Purpose: Packaging → Product

- Corners double as **record display stands**
- Designed for **reuse**, not landfill



Audio Technica Core Values and User Experience

Our design is congruent with Audio Technica's core value of accessible, high-quality audio gear, **as sustainability should also be accessible to everyone.**

Our design creates the ideal user unboxing experience as it is durable, reusable, and visually appealing.

- Cardboard packaging ensures safe transport, sustainability, and reusability.
- The elegant design doubles as a vinyl record holder, reminding customers of their eco-friendly choice.

Bonus: Integrating Japanese Cultural Elements

- Replaced plastic cover with a **Japanese tenugui towel**.
- Inspired by **furoshiki** (traditional wrapping cloth) aesthetics.
- Celebrates Audio-Technica's **Japanese heritage**.
- Tenugui sourced from **Hightide Store DTL**



Thank You. Questions?

